

Predicting Football Match Outcomes Using Machine Learning and Probability Calibration

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Abstract

Football match results depend on a complex interaction between team performance and competitive dynamics. In this paper, we apply a series of machine learning methods and Isotonic Calibration to predict match outcomes using a dataset of around 230,000 football matches. Bookmaker odds were adjusted with the Newton–Raphson method to account for bookmaker margin effects, while additional calibration procedures were used to improve probability estimation. The proposed machine learning models achieved an AUC of 70.71% in predicting final match outcomes (1-X-2), with particularly strong performance in away-win predictions, reaching an accuracy of 72.02%. Compared to traditional scoring approaches such as Over/Under goals, the proposed method produced more consistent results across different leagues. The analysis also showed that combining statistical match features with calibrated machine learning methods can improve prediction reliability in highly competitive environments.

Keywords: Machine learning; neural networks; stochastic techniques.

1. Introduction

Many real world problems can be tackled by the so - called machine learning tools. Among these problems one can find cases derived from various scientific fields, such as physics [1,2], astronomy [3,4], chemistry [5,6], medicine [7,8], economics [9,10], image processing [11], time series forecasting [12] etc. Common machine learning tools are artificial neural networks [13,14], decision trees [15,16], support vector machines (SVM) [17,18] etc.

In this research study, we address the problem of predicting the outcome of a football match by examining it through a variety of machine learning techniques. This problem was tackled by various researches during the past years.

Why is important???

For example, Hvattum et al. proposed the incorporation of ELO ratings for the prediction of match outcomes in their study [19]. Also, Owramipur et al. suggested the usage of Bayesian networks [20] to predict the outcomes for the Spanish League Barcelona team [21]. Similarly, Presetio et al. proposed the incorporation of logistic regression [22] to predict the outcomes of football matches [23]. Moreover, Martins et al. used a polynomial classifier for the prediction of football outcomes [24]. Schauburger et al. selected the method

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of random forests to predict the results of football matches in their study [25]. Various attributes of football players was studied by Danisk et al. in order to predict the result of football matches in their related paper [26]. Constantinou proposed a new machine learning model, named Dolores which is based on Hybrid Bayesian Networks, to predict football match outcomes from datasets selected in a series of countries [27]. Baboota et al. applied gradient boosting [28] for English Premier League [29]. Additionally, a deep learning framework was designed from Rahman to tackle the problem of prediction football results [30].

What we proposed here?? With references!

The remaining of this article is organized as follows: in section 2 the proposed datasets and the used methodology are fully described, in section 3 the experimental results are outlined and finally in section 4 a discussion is provided based on the experimental results.

2. Materials and Methods

2.1. The Dataset Employed

In this study, we utilized a large-scale dataset of historical football matches sourced from a professional sports analytics repository. The data collection process prioritized high granularity, incorporating not only match results but also advanced performance indicators and pre-match market probabilities. The dataset's reliability is ensured through its internal consistency and widespread use in sports forecasting benchmarks.

2.2. Dataset Description

The experimental dataset comprises 225,474 match events spanning from 2018 to 2026. To ensure rigorous evaluation, the data was partitioned into a training set of 149,998 matches and a dedicated test set of 75,476 matches. The core variables included in the raw data are summarized in Table 1. The inclusion of *LeagueId* and *SeasonPhase* allows the model to account for competitive context and seasonal variations.

Table 1. Description of the original dataset features.

Feature	Description / Range
MatchDate	Temporal identifier (2018–2026)
League / LeagueId	Categorical identifiers for competitive tiers
HomeTeam / AwayTeam	Competing squad identifiers
Label	Primary outcome target (1, X, 2)
EloDiff	Difference in Elo ratings (−500 to +500)
ProbHome/Draw/Away	Pre-match market probabilities (0 to 1)
RankDifference	Difference in league table standings
SeasonPhase	Categorical marker for season stage
AttackVsDefense / DefenseVsAttack	Derived offensive and defensive interaction ratios
TotalScoringPotential	Normalized long-term scoring capacity
TotalDefensiveWeakness	Normalized long-term goal concession rate
HomeCleanSheetRatio	Probabilistic historical clean sheet frequency
AwayFTSRatio	Away team Failed-to-Score probability

2.3. Feature Engineering and Mathematical Formulation

To capture the dynamic interaction between offensive and defensive capacities, the raw data was transformed into formal performance metrics. Let $AvgH_s$ and $AvgH_c$ denote the average goals scored and conceded by the home team, and $AvgA_s$, $AvgA_c$ the corresponding values for the away team. A smoothing constant $\epsilon = 0.01$ was applied to prevent division by zero.

AttackVsDefense: Represents the home team's offensive efficiency relative to the away team's defensive record:

$$\text{AttackVsDefense} = \frac{\text{Avg}H_s}{\text{Avg}A_c + \epsilon} \quad (1)$$

DefenseVsAttack: Quantifies the home team's defensive capability against the away team's scoring force:

$$\text{DefenseVsAttack} = \frac{\text{Avg}H_c}{\text{Avg}A_s + \epsilon} \quad (2)$$

TotalScoringPotential: Modeled as the numerical difference between home and away scoring capacities, identifying the relative offensive edge:

$$\text{TotalScoringPotential} = \text{Avg}H_s - \text{Avg}A_s \quad (3)$$

TotalDefensiveWeakness: Assesses the relative vulnerability of the home defense compared to the away defense as a ratio:

$$\text{TotalDefensiveWeakness} = \frac{\text{Avg}H_c}{\text{Avg}A_c + \epsilon} \quad (4)$$

HomeCleanSheetRatio (HCSR) and **AwayFTSRatio** (AFTS) capture probabilistic historical frequencies:

$$\text{HCSR} = \frac{\sum_{i=1}^N \mathbb{I}(G_{H,c} = 0)}{N}, \quad \text{AFTS} = \frac{\sum_{j=1}^N \mathbb{I}(G_{A,s} = 0)}{N} \quad (5)$$

2.4. Feature Importance Motivation

The generated feature space is explicitly designed to satisfy three essential, complementary operational criteria: (i) long-term structural team performance using rating systems (*EloDiff*, *RankDifference*), (ii) granular tactical interaction effects (*AttackVsDefense*, *DefenseVsAttack*), and (iii) distribution-based prior probabilities derived from betting market evaluations (*ProbHome*, *ProbDraw*, *ProbAway*). This multi-source configuration mitigates model vulnerability to single-feature bias and improves generalization accuracy.

3. Results

4. Conclusions

Author Contributions: M.S. and I.G.T. conducted the experiments, employing several datasets and provided the comparative experiments. M.S. and G.G. performed the statistical analysis and prepared the manuscript. All authors have read and agreed to the published version of the manuscript.

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